Why Kernel Development Matters

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A kernel is the core component of an operating system. It manages memory, hardware devices, process scheduling, interrupts, and communication between software and hardware.

Learning kernel development provides a deeper understanding of how computers actually work. Unlike application programming, kernel programming requires direct interaction with hardware and careful resource management.

The KFS project introduces several important concepts:

- Boot process and GRUB
- Assembly language entry points
- Memory layout and linker scripts
- VGA text mode output
- Low-level debugging and system architecture

These skills are valuable for systems programming, embedded systems, cybersecurity, virtualization, operating systems research, and performance engineering.